Last Resort Hotels — Draft Analysis Report

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1. Scenario Overview

Last Resort Hotels is an expanding hotel chain managing multiple hotel complexes, each containing sleeping accommodations, meeting spaces, recreational facilities, and guest services. As the organization grows, it requires a centralized database system to manage room availability, reservations, guests, events, billing, and facility usage.

Our proposed database consists of 33 interconnected tables designed to represent these operations. The schema prioritizes the hotel's physical structure, reservation management, customer relationships, event scheduling, and billing, while also incorporating room maintenance, access tracking, and guest communication.

2. Database Organization and Design Decisions

Hotel Structure and Rooms

The physical structure of the hotels is represented through hotel_complex, building, wing, floor, and room. This hierarchy allows rooms to be identified by their location within a particular hotel complex, since room numbers are not necessarily unique across the entire chain.

The room table contains information about room categories, capacities, amenities, rental rates, and whether rooms support sleeping or meeting activities. We use a common room entity rather than separate tables for each room type because certain rooms, such as suites, may serve multiple functions.

The room table also includes canSmoke, which indicates whether smoking is permitted in an individual room. This complements the smoking policy recorded for each floor and allows the database to distinguish smoking and nonsmoking rooms even on mixed floors.

Additional tables include bed_type and room_bed for recording bed configurations, room_connection for representing adjacency and connecting doors, and room_status_history for tracking cleaning, maintenance, and other changes in room availability.

Reservations and Assignments

The reservation system separates customer requests from specific room assignments. The reservation table stores booking details, dates, statuses, and deposits, while reservation_requirement and requirement_bed record preferences such as sleeping capacity, bed type, accessibility, and smoking restrictions.

The room_assignment table connects reservation requirements to specific rooms and records scheduled and actual occupancy times. This distinction is important because the scenario specifies that rooms may not be assigned until shortly before a reservation begins, and assignments may change due to maintenance or customer requests.

The assignment_guest table allows multiple guests to be associated with one room assignment and supports guests moving between rooms during their visits.

Customers, Organizations, and Events

We use a general party entity to represent both individuals and organizations. The person and organization tables store their respective information, while organization_person records organizational affiliations and authorized representatives.

This structure accommodates customers acting in different roles, including guests, event hosts, booking parties, and parties responsible for payment.

Events are represented through event, event_attendee, and event_room. These tables track event schedules, hosts, participants, and room usage. The time_slot table provides standardized meeting-room usage periods, while facility represents additional hotel facilities such as restaurants and recreational spaces.

Billing and Services

Billing is organized through service, charge, invoice, invoice_charge, and payment.

The service table defines available services and their default prices, while charge records individual expenses. The invoice table identifies the responsible billing party, and invoice_charge allows charges to be allocated across invoices. This supports split billing, including situations where an organization pays for accommodations while guests pay for personal expenses.

Payments are recorded separately from invoices, allowing the system to track outstanding balances and multiple payments toward a bill.

Access Tracking and Communication

The key_card, card_reader, and access_log tables represent the hotel's card-based access system. They record card assignments, reader locations, and entry or exit activity.

The staff and guest_message tables support staff responsibilities and guest communications, including confidentiality preferences.

3. Assumptions and Interpretations

The hotel description contains several ambiguities, so we made the following assumptions when designing the database:

- **Room functionality:** Sleeping rooms, meeting rooms, and suites can be represented through a common room entity with attributes indicating their capabilities. This accommodates rooms that serve multiple functions.
- **Smoking policies:** Each room has an individual smoking designation through canSmoke. Rooms must also comply with applicable floor-level restrictions, so rooms on entirely nonsmoking floors cannot permit smoking.
- **Reservation assignments:** Reservations may exist without specific assigned rooms. Assignments can be created or changed later, depending on availability and customer requirements.
- **Guest accommodations:** Multiple guests may occupy the same room, and individual guests may change rooms during their stay without requiring new personal records.
- **Customer roles:** Individuals and organizations may act as booking parties, hosts, or billing parties. A single party may occupy several roles.
- **Billing responsibilities:** Charges may be divided among multiple responsible parties, provided the total amount is accounted for. Meeting-room charges must have an identifiable responsible billing party.
- **Room availability:** Availability depends on both existing reservations and operational conditions, such as maintenance or cleaning.
- **Meeting-room scheduling:** Meeting rooms use designated time slots, while overnight accommodations use scheduled check-in and check-out timestamps.

4. Normalization and Relational Design

Our database is designed to follow Third Normal Form (3NF). Each table has a primary key, and relationships between entities are represented using foreign keys.

Many-to-many relationships are handled using intermediate tables. For example, room_bed associates rooms with bed types, event_attendee associates people with events, and invoice_charge associates individual charges with invoices.

We also separate related but distinct information to reduce redundancy. Reservations are stored separately from their requirements and room assignments, while charges, invoices, and payments

are maintained as independent entities. This helps prevent update anomalies and allows records to be changed without unnecessarily affecting unrelated information.

The final implementation will enforce appropriate uniqueness, referential integrity, and other constraints to maintain data consistency.

5. Scope, Tradeoffs, and Limitations

Although the schema accounts for the major requirements of the scenario, certain features have been simplified.

First, the database stores room characteristics and relationships but does not provide detailed graphical room layouts. Specialized configurations, such as movable walls, folding beds, and internal suite arrangements, are not individually modeled.

Second, the schema supports room assignments, customer preferences, and wing priority numbers, but automated room selection is not represented directly. Rules such as prioritizing the lowest-numbered available room, preventing overlapping assignments, and enforcing smoking restrictions would require additional constraints or application logic.

Third, complex pricing policies are not fully modeled. The scenario describes discounts for meeting-room usage, complimentary time slots, and surcharges for extended stays. Our database can record agreed rates and individual charges, but calculating these special prices would require additional business logic.

Fourth, customer qualifications used to determine deposits, such as payment history and flexibility, are not represented through a separate evaluation system. Instead, required deposits and amounts are stored directly with reservations.

Finally, the access-tracking system records card activity but cannot guarantee a guest's exact current location. Since certain card uses are optional, these records provide only partial information about guest movements.

These decisions allow us to focus on the central hotel operations while leaving room for future expansion.

6. Conclusion

Our proposed database provides a structured and extensible model of Last Resort Hotels' operations. By organizing hotel resources, reservations, customers, events, billing, and supporting services into related entities, the schema supports consistent data management while minimizing unnecessary duplication.

The design incorporates the most important requirements of the scenario and documents assumptions where the description is ambiguous. It provides a foundation for implementing the database and developing queries that analyze room usage, reservations, events, revenue, and other aspects of hotel operations.